WORKED SOLUTION

STEP II 2024, Question 3

Pure • Half-Angle Tangent Substitution

STEP II 2024, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Part (i) sets up a dictionary: the point at angle ϕ on the circle corresponds to the height $\tan\frac{1}{2}\phi$ where the line from N through it crosses the y-axis. Run it backwards too, so that every height q gives exactly one point on the circle. Then each function of q in the question becomes a trigonometric identity in $\frac{1}{2}\theta$, and we can read off what happens to the angle θ, and so to the point.

Solution

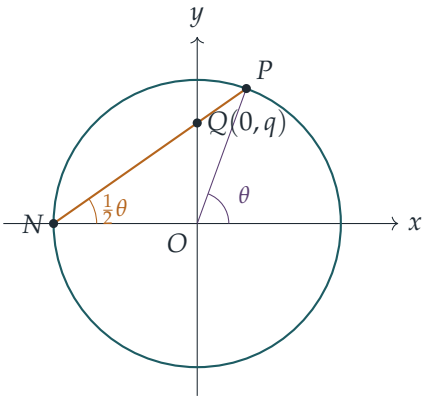
Part (i)

Since $P \neq N$, we have $\cos\theta \neq -1$, so the line NP is not vertical. Its gradient is

$$\frac{\sin\theta - 0}{\cos\theta - (-1)} = \frac{2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta}{2\cos^2\frac{1}{2}\theta} = \tan\tfrac{1}{2}\theta,$$

using the double-angle formulas $\sin\theta = 2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta$ and $1+\cos\theta = 2\cos^2\frac{1}{2}\theta$ (and $\cos\frac{1}{2}\theta \neq 0$, since $|\frac{1}{2}\theta| < \frac{1}{2}\pi$). The line passes through $(-1,0)$, so it is $y = (x+1)\tan\frac{1}{2}\theta$, and putting $x = 0$ gives $q = \tan\frac{1}{2}\theta$, as required.

Geometrically, this is the angle at the circumference: the chord NP makes angle $\frac{1}{2}\theta$ with the x-axis at N because P subtends angle θ at the centre.



Going backwards: from a height to a point

We will keep needing the converse. Given any real q, the line through N and $(0,q)$ is $y = q(x+1)$. Substituting into $x^2+y^2=1$:

$$x^2 - 1 + q^2(x+1)^2 = 0 \iff (x+1)\left[(1+q^2)x - (1-q^2)\right] = 0.$$

The root $x=-1$ is N, so the other intersection is

$$\left(\frac{1-q^2}{1+q^2},\ \frac{2q}{1+q^2}\right).$$

If $q = \tan\frac{1}{2}\phi$ then by the t-formulas this point is $(\cos\phi, \sin\phi)$. Since $\phi \mapsto \tan\frac{1}{2}\phi$ is a bijection from $(-\pi,\pi)$ to $\mathbb{R}$, we get the following.

Dictionary. If $\tan\frac{1}{2}\phi = q$ for *any* angle ϕ (not necessarily in $(-\pi,\pi)$), then the line from N through $(0,q)$ meets the circle again at $(\cos\phi, \sin\phi)$.

This holds for every ϕ because replacing ϕ by $\phi \pm 2\pi$ changes neither $\tan\frac{1}{2}\phi$ (which has period π in $\frac{1}{2}\phi$) nor the point.

Part (ii)

(a) Since $\tan\frac{1}{4}\pi = 1$, the addition formula gives

$$\tan\left(\tfrac{1}{2}\theta + \tfrac{1}{4}\pi\right) = \frac{\tan\frac{1}{2}\theta + 1}{1 - \tan\frac{1}{2}\theta} = \frac{1+q}{1-q} = \mathrm{f}_1(q),$$

which is $\tan\frac{1}{2}\left(\theta + \frac{1}{2}\pi\right)$, as required. The condition $q \neq 1$ means $\theta \neq \frac{1}{2}\pi$, so the denominator is non-zero.

(b) By (a), the height of Q_1 is $\tan\frac{1}{2}\phi$ with $\phi = \theta + \frac{1}{2}\pi$. By the dictionary, $P_1 = \left(\cos(\theta+\frac{1}{2}\pi), \sin(\theta+\frac{1}{2}\pi)\right)$. So

P_1 is the image of P under an anticlockwise rotation about O through $\frac{1}{2}\pi$.

(The excluded value $q=1$ is $\theta = \frac{1}{2}\pi$, the one point whose image under this rotation is N itself.)

Part (iii)

(a) P_2 is the point at angle $\theta + \frac{1}{3}\pi$, so by part (i), applied with the angle reduced into $(-\pi, \pi)$ if necessary (which doesn't change tan of half the angle), and $\tan\frac{1}{6}\pi = \frac{1}{\sqrt{3}}$:

$$\mathrm{f}_2(q) = \tan\left(\tfrac{1}{2}\theta + \tfrac{1}{6}\pi\right) = \frac{q + \frac{1}{\sqrt{3}}}{1 - \frac{q}{\sqrt{3}}} = \frac{\sqrt{3}q + 1}{\sqrt{3} - q}.$$

The condition $q \neq \sqrt{3}$ is $\theta \neq \frac{2}{3}\pi$, which is exactly when P_2 would be N and the line NP_2 wouldn't exist.

(b) This time use $\tan\frac{1}{4}\pi = 1$ with a difference:

$$\mathrm{f}_3(q) = \frac{1-q}{1+q} = \frac{\tan\frac{1}{4}\pi - \tan\frac{1}{2}\theta}{1 + \tan\frac{1}{4}\pi\tan\frac{1}{2}\theta} = \tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}\theta\right) = \tan\tfrac{1}{2}\left(\tfrac{1}{2}\pi - \theta\right).$$

By the dictionary, P_3 is the point at angle $\frac{1}{2}\pi - \theta$, namely $(\sin\theta, \cos\theta)$. The coordinates of P are swapped, so

$$P_3 \text{ is the reflection of } P \text{ in the line } y = x.$$

(c)

Ask yourself What does f_2^{-1} do to the angle?

From (a), f_2 sends $\tan\frac{1}{2}\phi$ to $\tan\frac{1}{2}(\phi + \frac{1}{3}\pi)$ for every angle ϕ (the calculation never used $\phi = \theta$). Hence $\mathrm{f}_2\big(\tan\frac{1}{2}(\psi - \frac{1}{3}\pi)\big) = \tan\frac{1}{2}\psi$, and since f_2 is one-to-one (solving $s = \mathrm{f}_2(q)$ gives the unique $q = \frac{\sqrt{3}s-1}{s+\sqrt{3}}$, for $s \neq -\sqrt{3}$),

$$\mathrm{f}_2^{-1}\big(\tan\tfrac{1}{2}\psi\big) = \tan\tfrac{1}{2}\left(\psi - \tfrac{1}{3}\pi\right).$$

So f_2^{-1} rotates the angle back by $\frac{1}{3}\pi$.

Now follow the angle through the three steps. As $0 < q < 1$, we have $0 < \theta < \frac{1}{2}\pi$.

- f_2: angle $\theta + \frac{1}{3}\pi \in (\frac{1}{3}\pi, \frac{5}{6}\pi)$. Here $\mathrm{f}_2(q) > 0$, so f_3 can be applied.
- f_3: angle $\frac{1}{2}\pi - (\theta + \frac{1}{3}\pi) = \frac{1}{6}\pi - \theta \in (-\frac{1}{3}\pi, \frac{1}{6}\pi)$. This is not $-\frac{2}{3}\pi$, so the height is not $-\sqrt{3}$ and f_2^{-1} can be applied.
- f_2^{-1}: angle $\frac{1}{6}\pi - \theta - \frac{1}{3}\pi = -\theta - \frac{1}{6}\pi$.

So $\mathrm{f}_4(q) = \tan\frac{1}{2}\big(-\theta - \frac{1}{6}\pi\big)$, and by the dictionary P_4 is the point at angle $-\frac{1}{6}\pi - \theta$.

Reflection in the line through O at angle α sends the point at angle θ to the point at angle $2\alpha - \theta$ (the mirror line bisects the two directions). Here $2\alpha = -\frac{1}{6}\pi$, so

P_4 is the reflection of P in the line through O at angle $-\frac{1}{12}\pi$ to the positive x-axis,

that is, the line $y = -\tan\frac{1}{12}\pi\, x = -(2-\sqrt{3})\, x$.

This makes sense: rotating by $\frac{1}{3}\pi$, reflecting in $y = x$ and rotating back is a reflection in the mirror $y = x$ rotated back by $\frac{1}{3}\pi$, and $\frac{1}{4}\pi - \frac{1}{3}\pi = -\frac{1}{12}\pi$. Explicitly, with $t = \tan\frac{1}{12}\pi = 2-\sqrt{3}$,

$$\mathrm{f}_4(q) = -\tan\left(\tfrac{1}{2}\theta + \tfrac{1}{12}\pi\right) = -\frac{q+2-\sqrt{3}}{1-(2-\sqrt{3})q}.$$

Checks

- Take $\theta = \frac{1}{2}\pi$ in (iii)(b): $P = (0,1)$, $q = 1$, $\mathrm{f}_3(1) = 0$, so $Q_3 = O$ and $P_3 = (1,0)$, which is indeed the reflection of $(0,1)$ in $y = x$.
- For (c) with $q = \frac{1}{2}$: $\mathrm{f}_2(\frac{1}{2}) = \frac{\sqrt{3}+2}{2\sqrt{3}-1} \approx 1.5146$, then $\mathrm{f}_3 \approx -0.2046$, then $\mathrm{f}_2^{-1} \approx -0.8868$, matching the closed form $-\frac{\frac{1}{2}+2-\sqrt{3}}{1-\frac{1}{2}(2-\sqrt{3})} \approx -0.8868$.
- A computer check over 100 000 random points P confirmed every relationship above (rotation by $\frac{1}{2}\pi$, the formula for f_2, reflection in $y = x$, and reflection in the line at angle $-\frac{1}{12}\pi$) to within rounding error.

Common slips

- Stating that P_1 is at angle $\theta + \frac{1}{2}\pi$ without noticing that this angle may lie outside $(-\pi, \pi)$; it is harmless only because $\tan\frac{1}{2}\phi$ and the point both repeat when ϕ changes by 2π.
- In (c), applying the rotation for f_2^{-1} in the wrong direction (adding $\frac{1}{3}\pi$ again), which gives angle $\frac{1}{2}\pi - \theta$ and so wrongly makes P_4 the same as P_3.

Beyond STEP

The map from the point at angle θ to $q = \tan\frac{1}{2}\theta$ is *stereographic projection* of the circle onto a line, and the functions f_1, f_2, f_3 are Möbius transformations $q \mapsto \frac{aq+b}{cq+d}$. Every rotation and reflection of the circle becomes such a transformation of the line, and composing the symmetries corresponds to multiplying 2×2 matrices $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Needham's *Visual Complex Analysis* develops this picture beautifully, extending it to the Riemann sphere.

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